

Engineering student building practical data, AI, and web tools. I turn messy real-world problems into usable workflows: data QA, modeling, validation, risk rules, APIs, databases, and deployable interfaces.

EDUCATION

National Cheng Kung University (NCKU)

B.S. student, Department of Hydraulic and Ocean Engineering

SELECTED PROJECTS

Buoy Data Analysis & Navigation Risk Platform

Built a Streamlit-based ocean buoy workflow with data-quality checks, LSTM time-series forecasting, train/validation evaluation, model caching, risk thresholds, and exportable outputs.

Python | Streamlit | Pandas | LSTM | Plotly

Chu River Chess — Online Chinese Chess

Built a browser-based Chinese chess product with 6-digit rooms, standard chess and Jieqi, AI mode, undo flow, replay, sandbox/custom setup, revision-based synchronization, and database-backed room state.

JavaScript | Next.js | Neon Postgres | REST API

NCKU Return OS — Credit & Course Decision Tool

Built an offline-first decision-support tool that tracks graduation requirements, prerequisites, timetable conflicts, risk scores, A/B/C plans, LocalStorage state, JSON backup, and CSV export.

HTML | CSS | JavaScript | LocalStorage

HIGHLIGHTS

- 2025 NODASS Ocean Big Data Competition - Great Prize.
- 2025 Qualcomm Taiwan AI Hackathon — 3rd place.
- 2025 Taiwan Ocean International Youth Forum — Leadership Star Award.

SKILLS

Data & AI	Python, Pandas, time-series modeling, LSTM, model validation, data-quality analysis, visualization
Web	HTML, CSS, JavaScript, Next.js, Streamlit, responsive UI, REST APIs
Backend & Tooling	Neon Postgres, Git/GitHub, GitHub Pages, LocalStorage, Node.js, CI/CD

以工程背景結合資料分析、AI 應用與 Web 產品開發，將真實資料與複雜規則整理成能操作、能驗證、能迭代的工具。

學歷

國立成功大學

水利及海洋工程學系 學士班

代表作品

浮標資料分析與航道風險評估平台

以 Streamlit 串接浮標資料品質檢查、LSTM 時間序列預測、訓練/驗證評估、模型快取、風險門檻判讀與輸出下載。

Python | Streamlit | Pandas | LSTM | Plotly

楚河棋局 | 線上中國象棋

完成 6 碼房間、標準象棋與揭棋、AI、悔棋、棋譜回放、沙盤、自訂棋局、revision 同步與資料庫持久化。

JavaScript | Next.js | Neon Postgres | REST API

NCKU Return OS | 學分地圖

把畢業學分、先修、衝堂、風險評分、A/B/C 修課方案、LocalStorage、JSON 備份與 CSV 匯出整合為決策工具。

HTML | CSS | JavaScript | LocalStorage

成果亮點

- 2025 NODASS 海洋大數據競賽 - 特優。
- 2025 高通台灣 AI 黑客松競賽 — 季軍。
- 2025 台灣海洋國際青年論壇 — 領航之星獎。

技能

Data & AI	Python、Pandas、時間序列、LSTM、模型驗證、資料品質檢查、視覺化
Web	HTML、CSS、JavaScript、Next.js、Streamlit、Responsive UI、REST API
Backend & Tooling	Neon Postgres、Git/GitHub、GitHub Pages、LocalStorage、Node.js、CI/CD